

Named vs positional

When you instantiate a module, you must connect its ports to nets in your design

Order matters for positional

- A D flip-flop might declare ports clk, d, q in that order
- Positional dff u open-paren clk, din, qout close-paren wires them correctly
- Swap the last two arguments to clk
- Named dot-d open-paren din close-paren dot-q open-paren qout close-paren makes the intent
- For a two-input AND gate, swapping a and b is harmless

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Named vs positional connections

Wire instances with `order` or with `.port(sig)`. Positional swaps fail silently — named connections make intent explicit.

Starter example: dff ports (clk, d, q) but positional instance is dff u(clk, qout, din) — **d/q swapped**. Fix with named .d(din), .q(qout).

Load starter example

Challenges (1/22)

Quiz: positional. Positional maps by? Answer: order

Show hint

Check

Next

Idle

quiz-pos

quiz-named

quiz-silent

quiz-prefer

starter

see-miswire

fix-pos

named-ok

named-code

and-swap

and-fix

explain

style-named

manual-fix

quiz-formal

quiz-actual

mod-and

preset-miswire-btn

clk-ok

quiz-reorder

both-ok

full-path

Core ideas

Positional

Nth actual maps to Nth formal — order is load-bearing.

Named

.port(sig) — order free; intent readable.

Hazard

Swapped positionals still compile; wrong wires.

Instance wiring

Connection sketch

Named vs positional lab starter

learn_verilog — named vs positional

Real Verilog practice

```
wsl - module10-named-vs-positional/examples/navigation

# Track A - real Unix
# real Verilog session (Track A)

$ cat examples/instance_conn/instance_conn.v
// instance_conn.v - prefer named port connections
module dff(
    input    clk,
    input    d,
    output reg q
);
    always @(posedge clk)
        q ≤ d;
endmodule

module top(
    input clk,
    input din,
    output qout
);
    dff u_good (
        .clk(clk),
        .d (din),
        .q (qout)
    );
    // dff u_bad(clk, qout, din); // positional swap - compiles, miswired
endmodule

$ iverilog -t null examples/instance_conn/instance_conn.v
(syntax check passed)
```

Real Verilog — named instance connections

Pitfalls to watch

- Do not assume positional order matches how you drew the schematic
- Do not mix named and positional on the same instance unless your style guide allows it
- Do not rely on the compiler to catch swapped functional ports
- For generated or macro cells with dozens of ports

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser
- On real Verilog, rewrite one instance using named ports
- When you are ready, take the short quiz, then continue to localparam in the next module